

Maurizio Ungaro

Scientific CV

Staff Scientist, Jefferson Lab

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Profile

Nuclear physicist and simulation software developer at Jefferson Lab working on Geant4-based detector simulations, [GEMC](#), CLAS12 production workflows, detector systems, and nucleon-structure research. My work connects physics analysis, detector operations, and simulation infrastructure, with a focus on making Geant4 workflows easier to build, run, document, and share.

Current Position

Staff Scientist

2011–present

[Jefferson Lab](#), Experimental Hall-B, Newport News, VA

Hall-B and CLAS12 detector operations, simulation software, Geant4 support, LTCC calibration and performance, distributed simulation production, and nucleon-structure analysis.

Previous Positions

Postdoctoral Research Associate and Research Associate

2004–2011

University of Connecticut, Storrs, CT

CLAS analyses, meson electro-production, detector software, and CLAS12 simulation and reconstruction development.

Education

2003 Ph.D. in Nuclear Physics, [Rensselaer Polytechnic Institute](#), Troy, NY. Thesis: π^0 electro-production from $\Delta(1232)$ at high momentum transfer.

1999 Laurea in Fisica, [Università degli Studi di Genova](#), Genoa, Italy.

Research and Technical Leadership

- Nucleon-structure measurements using meson electro-production and CLAS/CLAS12 data.
- Development and support of [GEMC](#), a database-driven Geant4 simulation workflow.
- CLAS12 simulation releases, geometry tags, and production workflows through [CLAS12 Simulations](#).
- Distributed simulation production using HTCondor and Open Science Grid resources.
- CLAS12 [Low Threshold Cherenkov Counter](#) operation, maintenance, calibration, and detector-performance studies.
- [Geant4 at JLab](#) tutorials, examples, and user support for Jefferson Lab users.

Selected Projects

GEMC: Database-driven Geant4 simulations with a Python-friendly workflow.

<https://gemc.github.io/home/>

Geant4 at JLab: Tutorials, examples, and support material for Jefferson Lab users.

<https://jeffersonlab.github.io/g4home/>

CLAS12 Simulations: Tagged CLAS12 geometry and simulation releases.

<https://github.com/gemc/clas12Tags>

CLAS12 on OSG: Distributed CLAS12 simulation production workflows.

<https://maureeungaro.github.io/home/osg/osg>

Low Threshold Cherenkov Counter: CLAS12 forward detector work for pion/kaon discrimination.

<https://maureeungaro.github.io/home/ltcc/ltcc>

Selected Collaborations and Facilities

- [Jefferson Lab](#) and [Jefferson Lab Hall-B](#): current laboratory and experimental program.
- [Heavy Photon Search](#): dark-sector search collaboration at Jefferson Lab.
- [SPRING-8](#): prior experimental work in Japan through LEPS/SPRING-8 collaboration.

Technical Skills

Simulation and analysis: Geant4, GEMC, CLAS12 simulations, ROOT, detector geometry, digitization workflows, PyVista visualization.

Programming and infrastructure: C++, Python, shell scripting, Git, GitHub, CI, Docker, Meson, CMake, SCons.

Scientific computing: HTCondor, Open Science Grid, SQLite, CSV/JSON/ROOT workflows, reproducible environments, documentation.

Communication: tutorials, technical writing, presentations, mentoring.

Selected Publications

- *Geant4 Monte-Carlo (GEMC): A database-driven simulation program*, EPJ Web of Conferences 295, 05005 (2024).
- *The CLAS12 Geant4 simulation*, Nucl. Instrum. Meth. A 959 (2020).
- *The CLAS12 Low Threshold Cherenkov detector*, Nucl. Instrum. Meth. A 957 (2020).
- *Measurement of the N to $\Delta(1232)$ transition at high momentum transfer by π^0 electro-production*, Phys. Rev. Lett. 97 (2006).

Online Profiles

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